

Computer Graphics

BCA — Semester 5 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Write the algorithm for Bresenham's line drawing? Digitize a line with end points A(6, 9) and B(10,5) using Bresenham's line drawing algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How can polygons be clipped? Describe Gouraud Shading model with necessary derivations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate between parallel and perspective projection. Perform the rotation of triangle with vertices A(5,7), B(12,85) and C(5,3) counterclockwise 30 degree about the pivot point (5,3).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain motion capture with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain architecture of random scan display systems with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain Cohen-Sutherland Line Clipping algorithm with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain about parametric curve. Describe the properties of Bezier curve.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Reflect a line segment having end points (9,3) and (12,10) about a line $X = 7$. Draw initial and final result graph as well.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain ambient light, diffuse reflection and specular reflection with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the wireframe representation of 3D objects.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How does a polygon can be created in OpenGL? Illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notesa. Z-Bufferb. Polygon Table

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.